Listen. Look. Correct.

One melody, one map

FIRULET MAP analyzes isolated guitar and estimates notes, timing, measured pitch and probable bends. Listen, correct the proposal and take MIDI into your DAW. Detection is approximate: use your ears as the reference.

Prepare your WAV

Isolated monophonic guitar: one note at a time, without accompaniment. Mono/stereo PCM16, PCM24 or float32 WAV, 44.1 or 48 kHz, up to 40 seconds. Conversion prepares an internal copy without changing your original WAV. A DI signal and BPM settings are not required for detection.

Suggested workflow

Open WAV → listen and review → save a .firulet → edit notes or reanalyze a section → compare Original/Edited → export MIDI. Audio and analysis operations run locally.

What changes in 0.7.0

Avalonia interface

FIRULET MAP moves to Avalonia on Windows x64. The analysis, editing, reanalysis and MIDI export workflows remain available. Press F1 for the shortcuts in this version.

Documents from earlier versions

You can open .firulet files from the previous version. Keep the original WAV with the document and use Relink WAV if it moved.

Installation and first analysis

Portable

Extract all of FiruletMap-0.7.0-win-x64.zip into a writable folder. Run FiruletMap.Desktop.exe and keep its DLLs and subfolders. .NET and Python are private; no global installation is needed.

Installer and update

Extract the installer ZIP and run setup. Administrator access is required; installation is for all users. Private .NET and Python runtimes are included. Close the app before updating. Updates are manual; uninstall through Windows Installed apps.

Open WAV / Analyze

Open WAV starts analysis automatically. Five stages show activity and elapsed time, not percentages or a guaranteed duration. Cancel analysis stops the work; Analyze repeats it. Record any Windows warning: the app and setup do not yet have their own code-signing certificate.

Save your analysis

Document and separate WAV

Save or Ctrl+S writes a .firulet document with analysis, edits, revisions, annotations and state. Save as creates another copy. The WAV remains separate: audio is not embedded. Keep both files when archiving or sharing. The icon beside Analysis copies the open document path.

Open and relink

Open analysis opens a .firulet without reanalysis. Ctrl+O opens a document or WAV. If audio is missing, use Relink WAV; its identity is checked and a different recording is rejected. Save after relinking. Opening another file or closing offers Save / Discard / Cancel for unsaved changes.

Keep the source

Original preserves the first analysis. Edited contains the effective result with your edits and accepted reanalysis. Saving never changes the audio. Multiple tabs and cloud synchronization are not included.

Navigation, playback and grid

Visual references

Waveform and map share time. Time zoom or the wheel over waveform/map zooms around the pointer. Pitch zoom changes semitone spacing and block/line size. Bar/seconds and pitch references stay fixed while scrolling. Full clip and Fit pitches help restore your view.

Audio, synth and cursor

Choose Original audio or the local synth. Foreground selects Original or Edited; synth plays only that layer. Show ghost overlays the other reference. Space plays/stops and returns to the starting position; Home goes to zero. The green curve is measured pitch and does not follow manual transposition.

Grid

Set quarter-note BPM, one fixed meter per WAV (including 27/16 or 23/16), Bar 1 origin and whole-to-sixteenth resolution. Valid changes enable the grid. Notes are not quantized. Show pitch rows and Show uncertain regions hide guides/shading without removing evidence.

Edit notes on the map

Select and change pitch

Choose Edited and close Section reanalysis to edit on the map. Click selects; Ctrl/Shift selects groups. Up/Down changes a semitone; Ctrl+Up/Down an octave. Right-click provides note/group/gap commands and dialogs for precise pitch or boundaries.

Create, delete and resize

Double-click a gap to create on that semitone row. Preferred length, initially 0.25 s, fits the available gap/audio; occupied time is rejected. Double-click a note or press Delete to remove it. Drag an edge with ↔ to change start/end; Esc cancels. Overlap and whole-note horizontal movement are not allowed.

Split, join and restore

S splits one note at its midpoint. J joins consecutive notes: confirm pitch and gaps; an incompatible bend may need removal. The context menu offers more options. Restore detected result restores detection in the confirmed interval and removes its corrections. Ctrl+Z/Y undoes/redoes; action history alone does not make a note Edited.

Confirmation and bends

Confirm what you heard

Under Edit notes → Confirmation and bends, Confirm notes marks selected uncertain notes as verified, individually or in a group. It also works after pitch/bend edits: Edited and Manually confirmed are separate facts. Remove confirmation reverses it. Engine evidence and diagnostic shading remain. Later musical changes invalidate affected confirmations.

Set / Edit

Select one note. Set creates an upward bend; Edit shows phases of an existing bend. Presets are ½ tone (1 semitone), Whole tone (2) and ½ semitone (0.5). Drag gold handles horizontally to adjust rise, hold and return. Release applies one Undo change; Esc cancels. No Apply is needed. Delete bend removes it.

Curves and limits

Green: measured pitch, independent of editing. Cyan dashed: manual bend. One bend per note; new bends rise by 0.5 to 2 semitones. Timing edits may retain the exact height of a larger detected bend. Replacing a detected shape asks for confirmation. Synth and MIDI use effective bends.

Annotations by time interval

Create and write

Double-click a gap in the Annotations lane above the waveform to create an annotation up to 1 second long, fitted to free space. Double-click an item to open Tools and select its title. Title appears on the bar; Description supports multiple lines. Apply commits text; Delete removes the annotation.

Move and resize

Drag the body until it reaches another item or an audio boundary; it never pushes other items. Drag an edge with ↔ to resize. Touching edges resize together and stay joined. There is one lane without overlap. Esc cancels; Ctrl+Z/Y undoes/redoes.

Text shortcuts

Enter in Title applies. Enter in Description inserts a new line. Ctrl+Enter applies from either field. Delete while the lane has focus removes only the selected annotation, not musical notes. Save the .firulet to preserve text, intervals and changes.

Section reanalysis

Select a section

Open Tools → Section reanalysis. Click and drag on waveform/map to draw a region; gold ↔ edges adjust it. Right-click offers an exact range or the selection. The cursor anchors to the section start. Replacement includes notes crossing boundaries; context is analyzed without replacement.

Compare before accepting

Choose Conservative, Balanced or Sensitive, then Reanalyze section. These are segmentation profiles of the same detector, not another algorithm or a guaranteed improvement. Compare Current and Proposal using Show proposal, ghost and synth. The yellow notice says the proposal is NOT applied and briefly pulses on outside clicks. Accept replaces the section in one Undo step; Discard keeps Current.

Mode and protection

While the panel is open, map gestures select regions. Close Section reanalysis to edit notes/bends again. Closing does not accept a pending proposal: Accept or Discard is still required. Save and MIDI use Current until Accept. Actual corrections/deletions are protected; restore detection if you want to discard them. Reverted history alone does not block analysis. Cancellation keeps Current.

MIDI into your DAW

File or drag

Tools → MIDI export: Export MIDI saves a file; Drag MIDI to DAW transfers it by dragging. Export the full analysis or enable a time fragment and select its range. Fragments rebase to zero and preserve relative rests. MIDI uses Edited/Current even when listening to Original or an unaccepted proposal.

Timing and bends

The file uses configured BPM/meter, 960 PPQ and ±12-semitone pitch-bend range. When importing into REAPER, honor MIDI tempo and use an instrument compatible with that range. There is no automatic quantization: pitches, boundaries and timing reflect edits and real performance. Bar 1 is a visual reference and does not shift exported notes.

Practical check

Compare a fragment with the original WAV; check pitch, rests, repeats and bends in the destination instrument. The built-in synth is a simple local reference, not the guitar timbre. REAPER and remote services are not required to use the app.

Shortcuts, limits and feedback

Keep help handy

F1 or ? opens Keyboard shortcuts. Ctrl+O: open; Ctrl+S: save. Space: play/stop and return; Home: zero. Up/Down: semitone; Ctrl+Up/Down: octave. S: split; J: join; Delete: remove. Ctrl+Z/Y: Undo/Redo outside text fields. Text fields and dropdowns retain normal keys; annotation shortcuts are described on their page.

Version limitations

Windows x64. No macOS, 96 kHz, recording, chords/polyphony, source separation, tempo maps, quantization or multiple tabs. Fast low notes, legato and tapping may fail. Reanalysis does not guarantee a perfect transcription. Clean-Windows testing, own code signing and automatic updates remain pending.

Tell us what happened

In the form, include 0.7.0, portable or installer, WAV format, steps, expected and observed behavior. Audio stays local; uploading it is not required. Record errors and keep the .firulet with its WAV. The form is in Spanish and contact details are optional.

Open feedback form